

1 Part 1: Bright Solitons in the Non-Linear Schrodinger Equation

Here we explore numerical simulations for interacting wave solutions via the non-linear Schrodinger equation. In 1 spatial dimension, we use,

$$\frac{\partial \psi}{\partial t} = i \left(\frac{\partial^2 \psi}{\partial x^2} - g |\psi|^2 \psi \right), \quad (1)$$

where $\psi(x, t)$ is the complex wave function and g is the interaction strength. We work here with attractive interactions which is where $g \leq 0$. This gives us "bright soliton" solutions which are stable wave packets. In this section, we explore their dynamics under varying conditions and with respect to 3 key quantities, interaction strength, velocity and relative phase.

1.1 A: Evolution Implementation

We aim to solve equation 1 on the domain $-L/2$ to $L/2$ where we set $L = 20$ and with Dirichlet boundary conditions $\psi(-L/2, t) = \psi(L/2, t) = 0$. We use RK4 as with assignment 1 and approximate the spatial derivative via the central difference method. We also set a grid size of $N = 128$, a time step of 0.01 and a max time of 20. We save data every 20 steps.

To implement evolution I used 2 separate functions. Firstly we define the calculateRHS function which finds the right hand side of equation 1 via the derivative term and the nonlinear term and combines them after multiplying by i . We also set our boundary conditions to 0 for the RHS as the time derivative of those points must be 0. Next we use the RungeKutta4 method to update the complex vector ψ using the RHS function to find the time derivative at the 4 times required.

1.2 B: Plane Wave

To solve for the evolution of a plane wave we must use the aforementioned functions alongside an initialisation function. We initialise the wavefunction as $\psi(x, 0) = \cos(\pi x/L)$. Note that we also normalise this after initialising to ensure it remains a physical quantum system.

We then evolve it via RK4 and calculate the wave amplitude. We loop over 5 values of g (-10, -3, -2, -1, -0.5, 0) which show a large range of interaction strengths. The evolutions are saved in txt files and then animated and saved as a gif in the Plots folder. We name it nls_animation_plane_wave.gif This shows each of the plane waves with different g values as they evolve. We can observe that they all remain stationary as expected but warp by becoming more localised at $x = 0$ and then spreading out periodically.

This is because under strong attraction $|g| \gg 1$ the system begins forming a bright soliton-like structure which resembles a localized blob of high amplitude around $x = 0$ and then spreads back out again.

More physically, the system prefers to minimise energy, and a strong attractive interaction lowers energy by clumping the wavefunction together at the centre of the plane wave. This results in a localized bound state — the hallmark of bright solitons. And noticably, the stronger the interaction, the tighter the localization.

For future questions we set $g = -1$.

1.3 C: Wave Packet

Next we investigate the dynamics of a wavepacket which is defined by the intial condition,

$$\psi(x, 0) = \sqrt{2} e^{iux} \operatorname{sech}(x). \quad (2)$$

We plot the dynamics as before as an animation for 5 values of u (-1, -0.5, 0, 0.5, 1) to give us both positive, negative and a 0 value. This is saved in the plots folder accordingly. We can clearly observe that for $u = 0$, the wavepacket stays stationary while for negative values it moves to the left and for positive values it moves to the right. This clearly indicates that u represents the velocity of the wavepacket. This is hence our bright soliton.

Additionally, we can observe that due to our Dirichlet boundary conditions, the soliton 'bounces' off the walls when it gets close. We can hence track the peak position of the wave amplitude over time for our values of u and observe the effects. We plot the resulting figure in 1. This clearly demonstrates the effect of u is to add a velocity to the wave packet.

For the next question we set $u = 0.1$.

1.4 D: Solitons

Now we expand our simulation to solve for the evolution of 2 solitons. To do this we simply change our initial condition to,

$$\psi(x, 0) = \sqrt{2} \exp(iux) \operatorname{sech}(x + L/4) + \sqrt{2} \exp(-iux + i\theta) \operatorname{sech}(x - L/4). \quad (3)$$

This equation represents 2 solitons beginning halfway on either side of the lattice, moving towards each other with velocity u and with a relative phase of θ . Again we explore the dynamics of the solution by animating their motion for 5 values of theta ($2 * \pi, \pi, 2.0 * \pi/3.0, \pi/2.0, 0$). We can clearly observe from the animation that for theta close to 0 and 2π the solitons merge while for others the solitons bounce, with the largest bounce occuring at $\theta = \pi$ which is when the solitons are most out

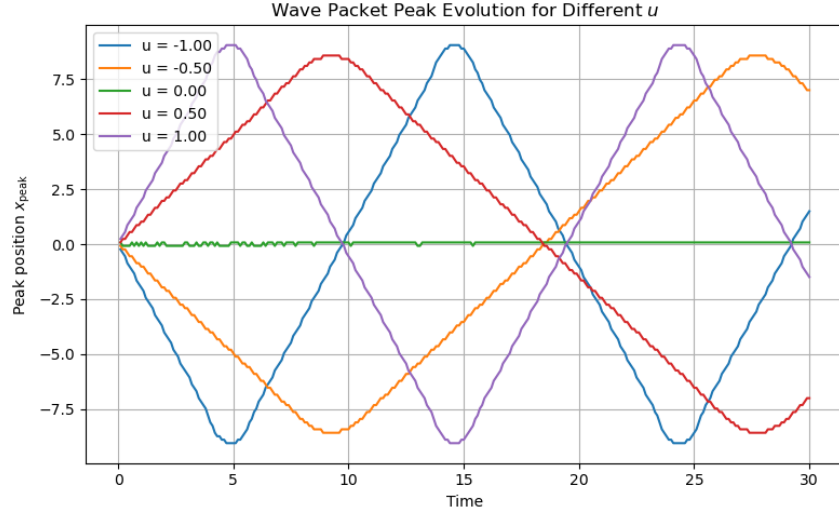


Figure 1: Plot of the peak position of the wavefunction over time for the wave packet initial condition. We can clearly observe that for $u = 0$, the packet does not move from the centre of the grid. Whereas for large u , the peak moves faster and bounces off the walls earlier. We also see symmetry around $u = 0$ indicating that the sign of u represents the packets velocity.

of phase. To investigate this further we can monitor the positions of the 2 peaks and the distance between them in figures 2 and 3.

This demonstrates what we observe qualitatively from the animation, where the solitons get closer and bounce/merge. We can also plot the distance of closest approach of the solitons against our values of θ to determine what their behaviour looks like and the dependence on θ . The results are presented in figure 4. Here we see the same behaviour as when the solitons are in phase, they get closer than when they are out of phase.

Using these observations we can hypothesise why the solitons appear to merge or bounce for different phases. It is clear that the relative phase influences how the solitons interfere when they overlap and we can observe that for $0, 2\pi$ they constructively interfere and hence 'merge' while for other values they destructively interfere (or a mix) and hence a single peak is not evident, instead bouncing off each other. Due to the nonlinearity in the NLSE, the constructive interference causes them to temporarily form a bound state, appearing to "merge" before separating again.

We can think of the soliton interaction as governed by a phase-dependent effective potential which is attractive for in phase and repulsive for out of phase. This is because the nonlinear interference term in the NLSE can be viewed as a coherence-mediated force — constructive coherence increases local intensity and tightens interaction, while destructive coherence disperses it. This is the interaction that causes the appearance of bouncing or merging.

This is also clear from the form of the initial wavefunction where the primary difference in the exponential term is the relative phase. Clearly $e^{i2\pi} = 1$ and so when θ is a multiple of 2π , the terms become equivalent and can combine/merge. Alternatively, when θ is an odd multiple of π , we get a destructive term via $e^{i\pi} = -1$, which causes them to bounce.

hamiltonian stuff as well?

1.5 Conclusion

To conclude, we numerically investigated the dynamics of bright solitons in the one-dimensional non-linear Schrödinger equation with attractive interactions. Starting from simple plane wave and wave packet initial conditions, we validated the emergence of stable, localized structures characteristic of bright solitons. By varying interaction strength, initial velocity, and relative phase between solitons, we demonstrated how each parameter affects the evolution and interaction of these nonlinear wave packets.

We observed how the interaction strength causes localisation of the solution and peaking. We also showed how altering u affects the velocity of the Soliton and the interaction of solitons with our boundary conditions (bouncing off with some distortion). Most notably, we observed phase-dependent soliton interactions, where solitons merge or repel based on their relative phase—revealing an effective coherence-mediated interaction potential. These findings highlight the rich dynamical behaviour that arises from the interplay between dispersion and nonlinearity, and the crucial role of phase in governing soliton interactions.

Finally, all code is available on [Github](#) and the animations are found in the plots folder. We omit them in this report for brevity and space limitations.

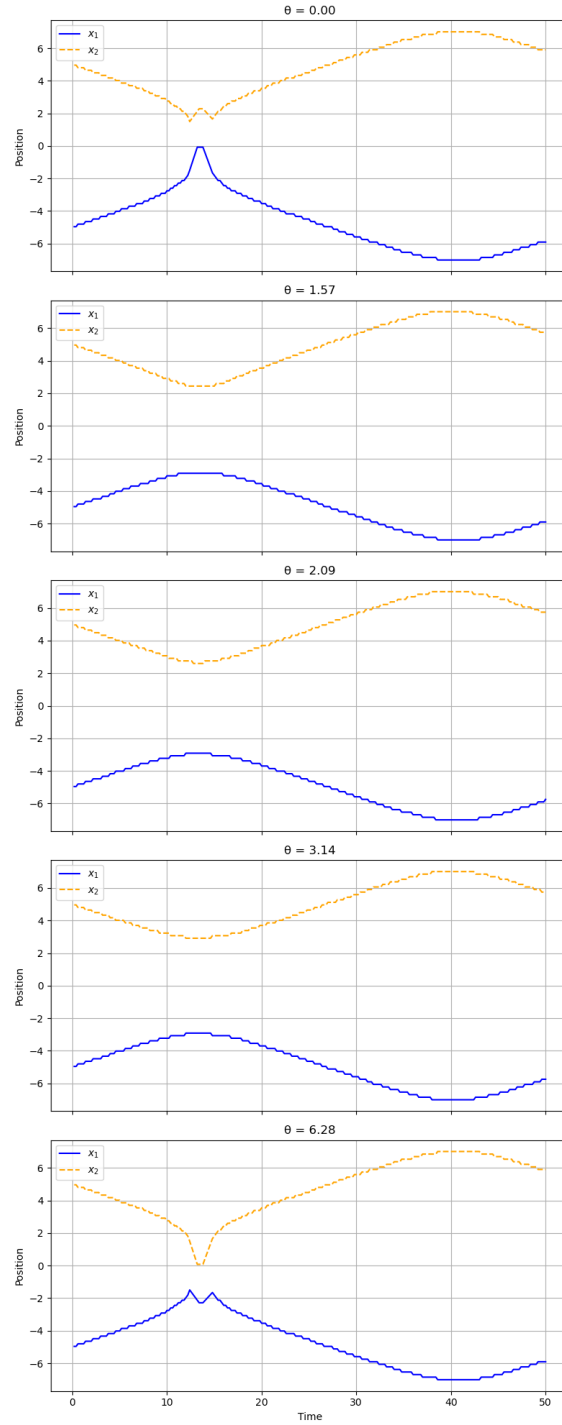


Figure 2: Plot of the peaks of the solitons over time for various relative phases, we can observe for $\theta = 0, 2\pi$ we have the closest peaks but we do not see evidence of the merge. This is likely an artifact from the method which we use to calculate the peaks. Regardless, we see the behaviour of 2 solitons moving towards each other and interacting differently depending on θ .

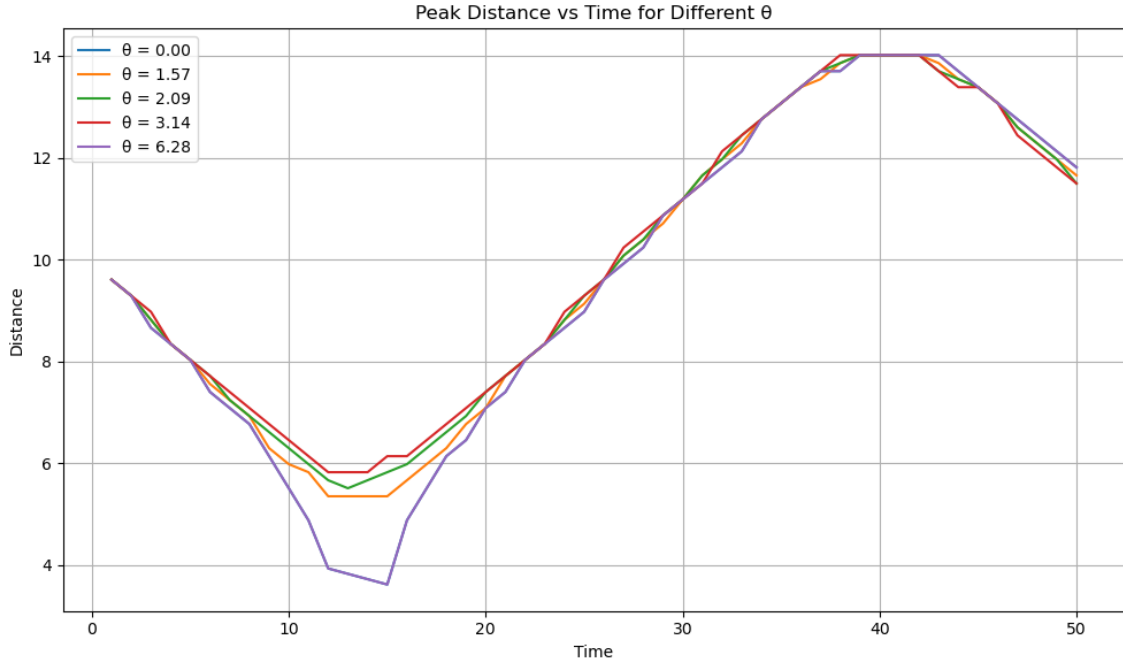


Figure 3: Plot of the distance between peaks vs time to demonstrate how the solitons interact and how close they get before merging or bouncing. We can clearly see the solitons moving towards each other before bouncing for off phase cases and then getting closer together for 0 and 2π . Note the lines for 0 and 2π are directly overlapping.

2 Part 2: The Transverse Ising Spin Model

Our next task is to investigate the transverse Ising spin model in 1 dimension to observe a quantum phase transition. Traditionally when we work with the standard Ising model, thermal phase transitions arise when symmetric properties of the equilibrium state change. After a critical point, the system becomes disordered, manifesting as a point of non-analyticity in the free energy as a function of temperature. In our case (many body quantum systems), an analogous effect occurs at 0 temperature. We can observe a point of non-analyticity in the ground state energy associated with a change in the symmetric properties of the ground state wave function's energy.

Note here that a sufficient condition for non-analyticity is that it has a point where the derivative is not defined. The 1D transverse Ising model is what we use to explore this, given by,

$$\hat{H} = -\omega \sum_{m=0}^{N-1} \hat{\sigma}_z^m - g \sum_{m=0}^{N-1} \hat{\sigma}_x^m \hat{\sigma}_x^{m+1}, \quad (4)$$

where,

$$\hat{\sigma}_z^{(m)} = I^{\otimes m} \otimes \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix} \otimes I^{\otimes (N-1-m)} \quad \text{and} \quad \hat{\sigma}_x^{(m)} = I^{\otimes m} \otimes \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \otimes I^{\otimes (N-1-m)} \quad (5)$$

are the z, x pauli spin matrices respectively for a single spin half particle located at the m th lattice site. I is the 2×2 identity, and $\otimes$ represents the tensor product. We impose periodic boundary conditions so

$$\hat{\sigma}_x^N = \hat{\sigma}_x^0. \quad (6)$$

The first term in the Hamiltonian describes the coupling of 1 atom to an external field that drives a transition between 2 single-particle states. The 2nd term describes the nearest-neighbour spin interaction with $g \geq 0$. This is analogous to the standard Ising model with a magnetic field and nearest neighbour interactions. We set $\omega = 1$.

We wish to explore the computational scaling of creating the Hamiltonian matrix as a function of N , how the ground state energy depends on g and hence if we observe a point of non-analyticity and finally how the dynamics of a quantum state depend on initial g and if we quench (rapidly change) g so the state is not longer in the ground state.

2.1 A: Execution Time

Our first task is to construct the Hamiltonian H . We begin by defining a Matrix class from prior assignments. We do not modify the data type as our matrices also still use real values, it is only our state vectors which will be complex. We add methods to extract elements from the matrix, print out the matrix and add matrices as standard operations. We also reuse our eigendecomposition code to find the eigenvalues and vectors using LAPACK. The primary novel addition for this matrix

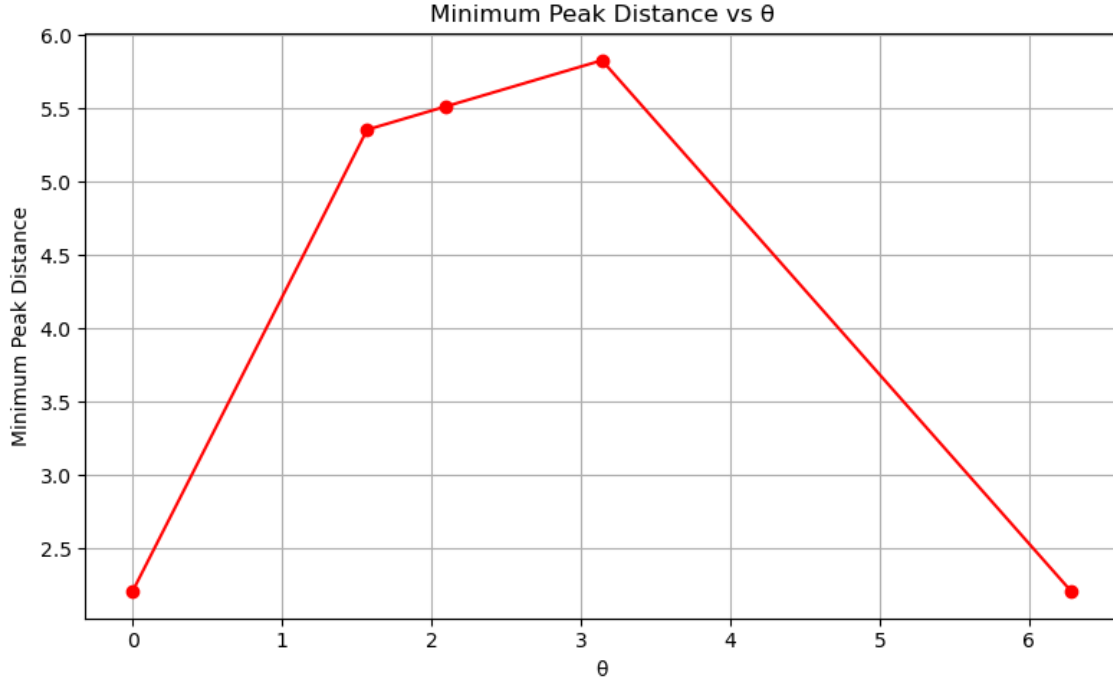


Figure 4: Minimum peak distance of the solitons for our 5 values of θ . We can clearly see when the relative phase is in phase, the solitons get closer than when the phase is out of phase. The dynamics are also symmetric around π as expected.

class is the inclusion of a tensor product method. This is a method we can call on a matrix from this class to tensor product it with another matrix (from the same class). To do this, we create a new matrix of the appropriate size (rows1 times rows2 and same for columns) and then iterate through a for loop where we find each element via elementwise multiplication. This is all found in the Matrix.hpp file.

Our next step is to find each term of the Hamiltonian, to do this we find the σ_z^m matrices and the $\sigma_x^m \sigma_x^{m+1}$ matrices. For the former, we iterate from 0 to $N - 1$ starting with a 1 and then tensor product the matrix with the Identity or Z depending on the index. Similarly, for the latter, we use the hint that,

$$\hat{\sigma}_x^{(m)} \hat{\sigma}_x^{(m+1)} = I^{\otimes m} \otimes \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \otimes I^{\otimes (N-2-m)}. \quad (7)$$

to iterate from 0 to $N - 1$ and times by I unless the index equals m or $m + 1$ with appropriate boundary conditions. We do so similarly for x and z matrices. Then to find the first term, we use a for loop over N to find each matrix for m and add them together, noting that $\omega = 1$. For the 2nd term, we do the same process with $m, m + 1$ and multiplying by g . These processes are found in part2_functions.hpp.

We then add these terms together and find the Hamiltonian, from which we can find the eigenvalues and eigenvectors. We time this process as we loop over values of N and plot our results in figure 5. We can clearly observe an exponential trend which is logical as the size of our matrix scales as 2^{2N} . We also observe the time taken for the first loop to initialise which often adds to the time taken for the first iteration. A similar trend was observed in the Ising model when we used OpenMP for multithreading.

2.2 B: Ground State Energy

Our next goal is to observe the quantum phase transition. We do this by finding the ground state energy, E_0 , and plotting E_0/N as well as $\frac{\partial^2 E_0}{\partial g^2}/N$ as function of g for $0 < g < 8$. We also wish to plot against the analytical result in the thermodynamic limit,

$$\lim_{N \rightarrow \infty} \left\{ \frac{\varepsilon_0}{N} \right\} = -\frac{1}{2\pi} (2 + g) E \left[\frac{8g}{(2 + g)^2} \right], \quad (8)$$

where

$$E[x] = \int_0^{\pi/2} d\theta \sqrt{1 - x \sin^2(\theta)}. \quad (9)$$

We find the second derivative of a function via the central difference method as used in prior work. To find the ground state energy, we find the Hamiltonian for different g and then take the first eigenvalue. We then store this as we loop over g .

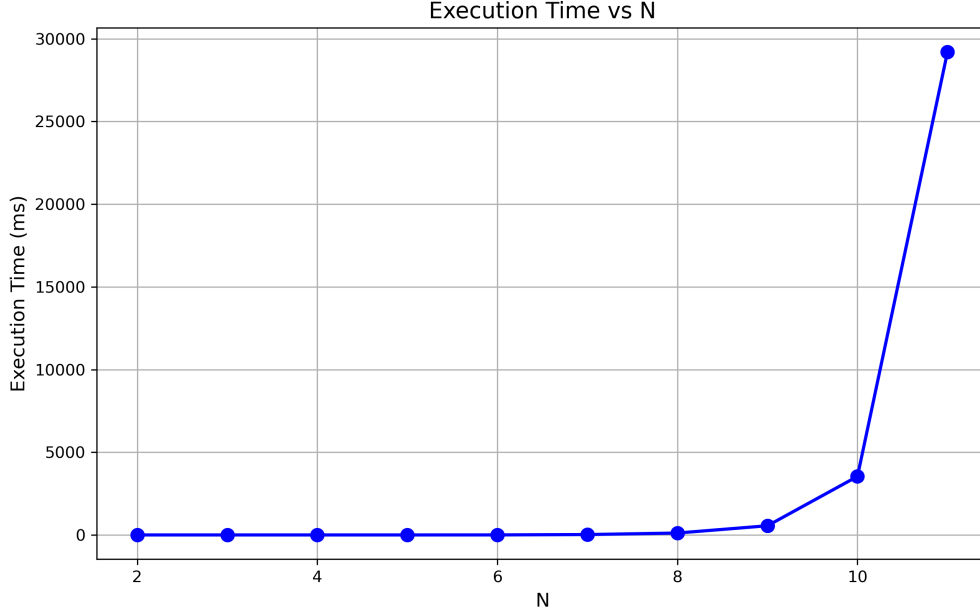


Figure 5: Plot of the execution time of constructing the Hamiltonian demonstrating an exponential time taken with a small overhead for the first iteration. This suggests the construction of the Hamiltonian scales logically indicating that the tensor product operation is likely functioning correctly.

To find the elliptical integral, we use a numerical approximation to the integral via simpsons rule for accuracy. Our results for the energy are shown in figure 6.

We clearly see an exact match for all g , indicating that the construction of the Hamiltonian is very accurate. We can see a sharp change in concavity around $g = 2$. This points to a location of non-analyticity which we investigate further via the 2nd derivative. This is shown in figure 7.

We clearly observe a point of non-analyticity in the theoretical energy values and a corresponding significant dip in the numerical values. We also see an almost exact match for most values of g except around the critical point. The discrepancy is likely because we are not at the thermodynamic limit where $N \rightarrow \infty$.

These results are clearly indicative of the fact we are observing a phase transition dependent on g . We can clearly see that the transition happens at $g = 2$, which is when $g = 2\omega$. This follows from our definition the Spin matrices as if we were to instead use pauli matrices with 1 in each element, the phase transition occurs at $g = \omega$. This demonstrates that when the interaction strength becomes stronger than the individual atom coupling, we observe a phase transition. This indicates that when $g > 2$, we enter a paramagnetic type regime, where

Explain when the transition happens and argue from the Hamiltonian what the phases are. What is the interpretation of the phase transition in the fermi representation and in momentum space?

2.3 C: Evolution of Observables

Our final task is to use this Hamiltonian to investigate the evolution of wavefunctions. To explore the non-equilibrium dynamics of a system, a common method is to start in the ground state of a Hamiltonian and quench (rapidly change) a parameter so it is no longer in the ground state. In our case, we do so with g . We investigate 10 cases split into 2 sections. Firstly we start with $g = 0$ and then evolve according to $g = 0, 1, 2, 3, 4$.

Computationally, we first find the ground state wavefunction via the first eigenvector associated with the lowest eigenvalue. We then implement RK4 by first finding the derivative, via $-iH|\psi\rangle$ using matrix vector multiplication. We then use RK4 and update the state vector accordingly. To observe dynamics we aim to track the time evolution of 3 observables, namely,

$$S_z = \sum_{m=0}^{N-1} \langle \hat{\sigma}_z^{(m)} \rangle, \quad (10)$$

$$S_x = \sum_{m=0}^{N-1} \langle \hat{\sigma}_x^{(m)} \rangle, \quad (11)$$

and

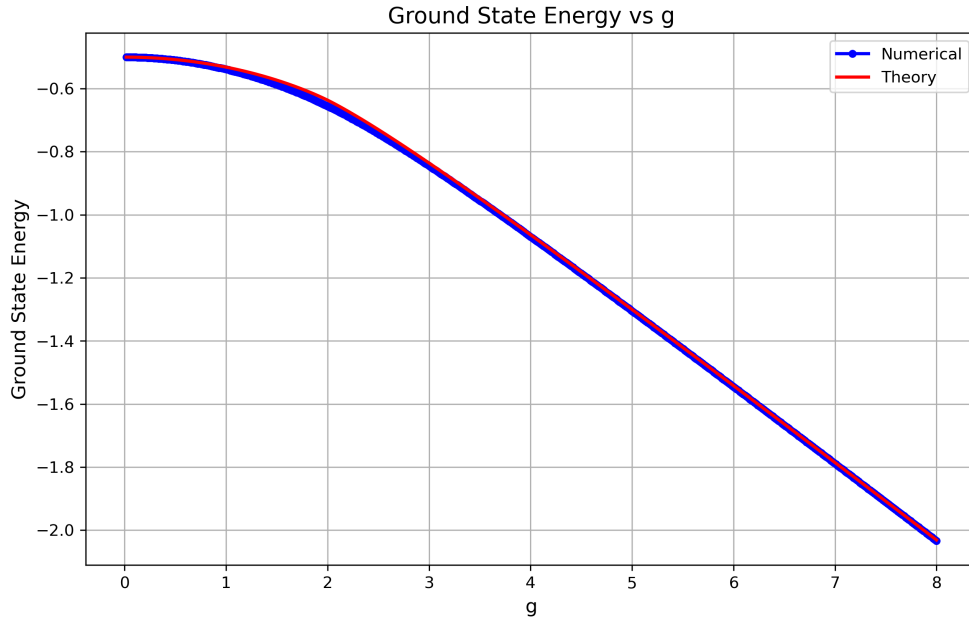


Figure 6: Plot of the ground state energy against g and with the theoretical values at the thermodynamic limit. We see an almost exact match and a noticeable change in concavity around $g = 2$, indicative of a point of non-analyticity. We also see a slight discrepancy around this point due to the fact we are not at the thermodynamic limit.

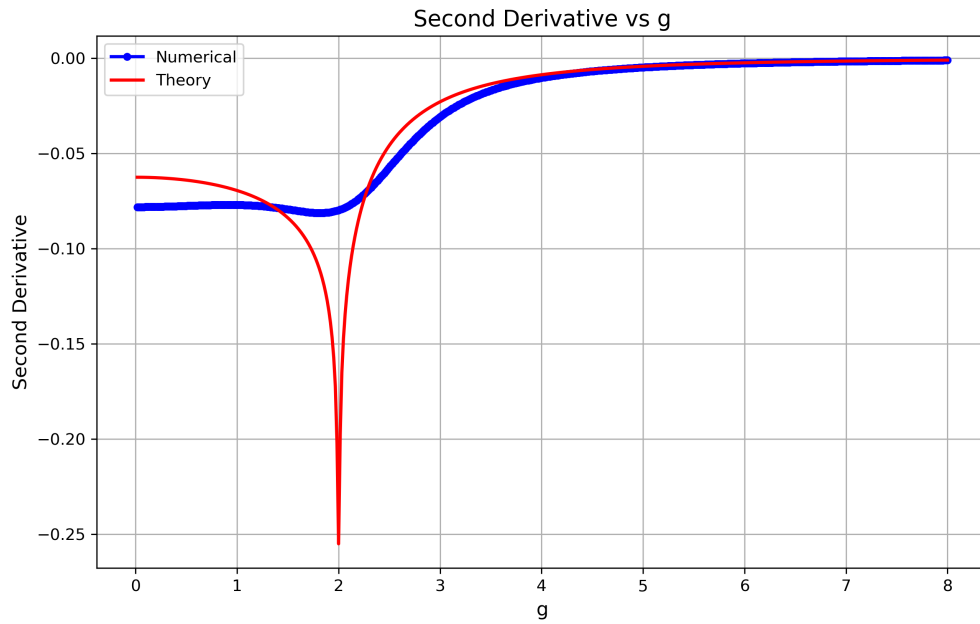


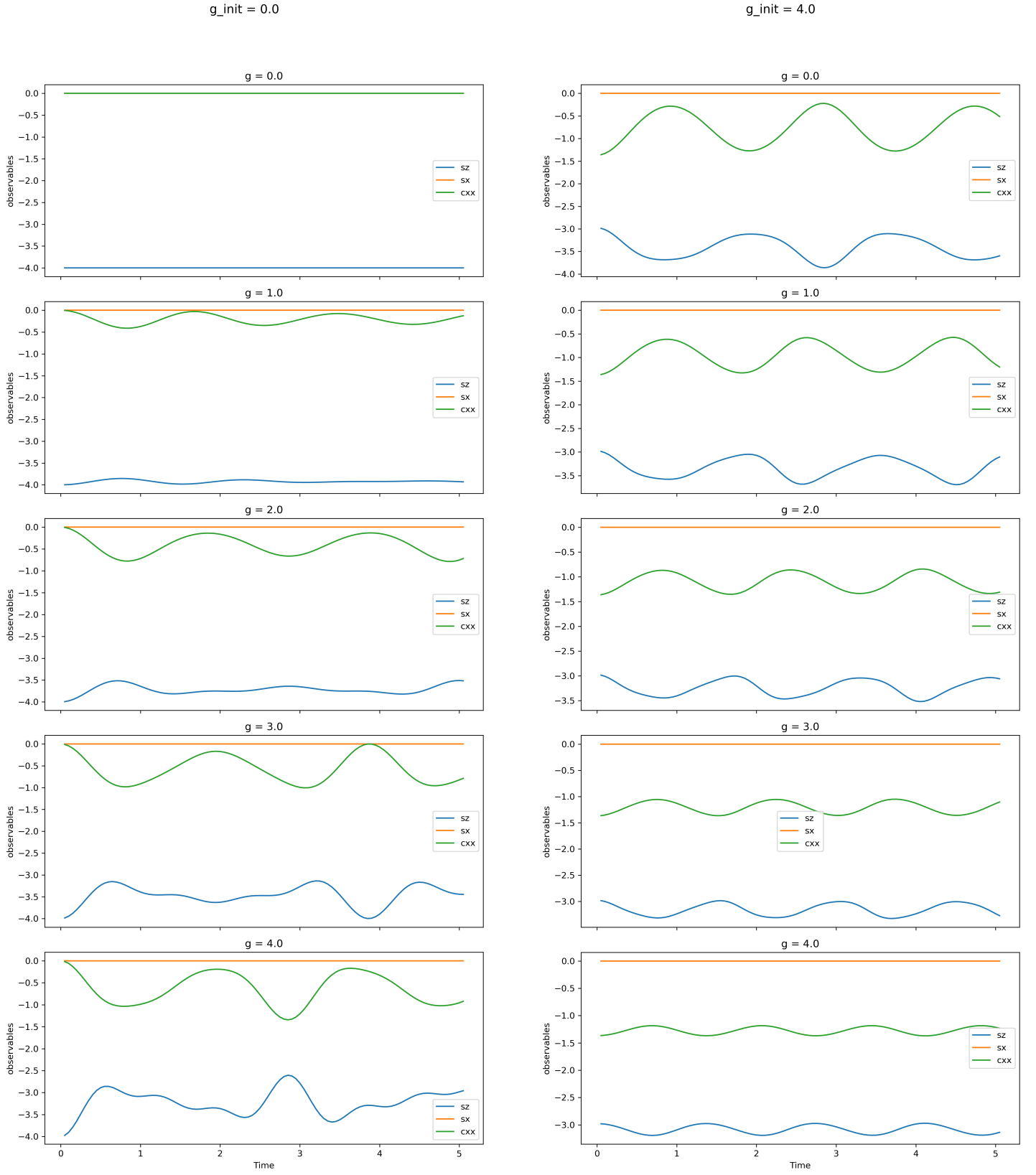
Figure 7: 2nd derivative of the ground state energy as a function of g with the theoretical and numerical data plotted together. We see evidence of the point of non-analyticity in the theoretical equation and a corresponding dip in the numerical simulations. This is indicative of a phase transition. Slight discrepancies are caused as we are not in the thermodynamic limit.

$$C_{xx} = \sum_{m=0}^{N-1} \sum_{\substack{n=0 \\ n \neq m}}^{N-1} \langle \hat{\sigma}_x^{(m)} \hat{\sigma}_x^{(n)} \rangle. \quad (12)$$

To calculate these, we find the σ matrices as before and then find the expectation value via the matrix vector multiplication method and the dot product method and sum over all m, n . We run the C++ code over 2 for loops for the initial g value and the evolution g values to produce the plots given in figures 8(a) and 8(b). We note that the results are sensitive based on the time steps chosen, the lattice size and other hyperparameters relevant to this problem. Specific choices for our results are shown in code and easily modifiable if required.

The salient features of these observables are firstly that S_x remains 0 in all cases regardless of initial g or evolution g .

2.4 Conclusion


 ((a)) Caption for $g_{\text{init}} = 0.00$

 ((b)) Caption for $g_{\text{init}} = 4.00$

Figure 8: Comparison of observables for different initial coupling values.